Rohith Karasala

Email: rohithkarasala6@gmail.com|
Phone: +1 (401) 285-1309
LinkedIn: linkedin.com/in/rohith-karasala

Professional Summary:

Senior Java Backend / Full Stack Engineer with **7+ years** of experience building scalable microservices, cloud-native applications, and enterprise software solutions using **Java**, **Spring Boot**, **React.js**, **AWS**, and **Python**. Strong background in **REST APIs**, **GraphQL**, **distributed systems**, **SQL/NoSQL databases**, **CI/CD**, **automated testing**, and **production support**. Experienced in delivering robust backend services and responsive front-end components while collaborating with cross-functional teams in Agile environments.

- Designed and developed high-performance backend services using **Java**, **Spring Boot**, and **Spring Framework** to support enterprise-scale applications.
- Built scalable **microservices architecture** to improve modularity, resilience, and service independence across distributed systems.
- Developed secure and efficient **RESTful APIs** for internal and external application integrations.
- Implemented **GraphQL APIs** with schema design, resolvers, queries, and mutations for flexible data access.
- Created and optimized **data processing pipelines** using **Apache Spark** for large-scale transformation and integration.
- Developed and orchestrated cloud-native workflows using **AWS Glue**, **AWS Step Functions**, **AWS Lambda**, **SNS**, **SQS**, and **S3**.
- Designed **event-driven architecture** using **Apache Kafka** for real-time communication and asynchronous processing.
- Secure engineering practices across **OAuth 2.0**, **IAM**, **PCI-DSS**, and **enterprise compliance requirements.**
- Ensured compliance with PCI-DSS and financial governance standards in payment, banking-adjacent, and insurance applications.
- Worked extensively with SQL and NoSQL databases including Oracle, MySQL, PostgreSQL, MongoDB, DynamoDB, DB2, and Teradata.
- Developed responsive front-end components using React.js, JavaScript, HTML5, CSS3, and Bootstrap for enterprise web applications.
- Contributed to full-stack feature development by integrating React-based user interfaces with Java backend services and APIs.
- Built and maintained CI/CD pipelines using Jenkins, Git, and Maven for continuous delivery and reliable release management.
- Containerized applications using Docker and supported cloud-native deployment practices for scalability and portability.
- Deployed microservices on AWS and supported modern cloud-based application development across distributed environments.
- Migrated legacy monolithic applications into **domain-driven microservices** to improve scalability and maintainability.
- Improved system performance and reliability by optimizing distributed architectures and reducing latency by approximately **30%**.
- Enhanced production observability through **Splunk**, **New Relic**, and **Application Insights**.
- Troubleshot and resolved production incidents, including **Sev2 issues**, using logs, telemetry, and root cause analysis.
- Participated in **on-call support rotations** to maintain system uptime and operational stability.
- Designed safe deployment strategies such as **blue-green deployments**, **canary releases**, and fault-tolerant rollout methods.
- Automated business and operational workflows using Java, Python, and UiPath to reduce manual effort and improve efficiency across support and release processes.
- Developed batch processing and scheduling solutions using **Spring Batch** and **Spring JDBC**.

- Implemented persistence layers using **Hibernate**, **JPA**, and **iBatis** for efficient database interaction.
- Used **HQL**, **Criteria API**, **Named Queries**, and **Hibernate interceptors** to improve data access and application behavior.
- Applied **Spring AOP**, **IoC**, and **Dependency Injection** to improve maintainability and cross-cutting concern management.
- Integrated asynchronous messaging using **JMS** and **Message-Driven Beans (MDBs)** for reliable inter-service communication.
- Worked across Agile environments, participating in sprint planning, backlog refinement, peer reviews, testing, and collaborative delivery using Scrum practices.
- Followed secure coding practices and encryption standards to protect sensitive financial and cardholder data.
- Collaborated with **architecture, QA, DevOps, data engineering, compliance, and business teams** to deliver scalable, secure enterprise solutions.

Technical Skills:

Languages:	Java, Python, SQL, JavaScript.
Frameworks:	Spring Boot, Spring Framework, Spring Security, Spring Cloud, Spring AOP, Spring Batch, Spring MVC, Hibernate, JPA, iBatis.
APIs & Integration:	RESTful APIs, GraphQL, JAX-RS (Jersey), JAX-WS (SOAP), JSON, XML.
Cloud Platforms:	Amazon Web Services (Glue, Lambda, Step Functions, SNS, SQS, S3, DynamoDB, IAM), Cloud-Native Deployment Practices.
Data & Processing:	Apache Spark, ETL Pipelines, Spring Batch.
Messaging & Streaming:	Apache Kafka, JMS.
Databases:	MySQL, PostgreSQL, MongoDB, DynamoDB, DB2, Teradata, Azure SQL.
Frontend:	React.js, Backbone.js, HTML5, CSS3, Bootstrap.
DevOps & Tools:	Docker, Jenkins, Git, Maven, JIRA.
Monitoring & Logging:	Splunk, New Relic.
Security:	OAuth 2.0, PCI-DSS, IAM.
Testing:	Mockito, Selenium WebDriver, Postman, Unit Testing, Integration Testing.
AI & Productivity:	AI-assisted development workflows, debugging support, documentation, code generation.
Architecture:	Microservices, Event-Driven Architecture, Distributed Systems.

Professional Experience:

Client: Amazon Web Services, Seattle, USA. **Sep 2022 – April 2026**
Role: Software Development Engineer

Responsibilities:
- Designed and developed high-performance backend services and full-stack application features using Java, Spring Boot, and Spring Framework, supporting large-scale distributed systems and enterprise application development in AWS environments.
- Built scalable **data processing pipelines** using **Apache Spark** and implemented robust **ETL workflows** for large-scale data transformation and integration.
- Developed and orchestrated cloud-native data pipelines using AWS Glue, AWS Step Functions, AWS Lambda, Amazon SNS, Amazon SQS, and Amazon S3, supporting scalable backend processing.
- Architected and implemented **microservices-based systems** using **Spring Boot**, **Spring Security**, and **Spring Cloud**, ensuring scalability, fault tolerance, and service resilience.
- Designed and developed secure RESTful APIs and GraphQL APIs (schema design, resolvers, queries, mutations) for payment systems, billing integrations, and third-party services.
- Implemented **OAuth 2.0** and **AWS IAM-based authentication/authorization**, ensuring secure access control and compliance with **PCI-DSS standards** for handling sensitive financial data.
- Built **event-driven architectures** using **Apache Kafka**, **AWS Lambda**, **SNS**, and **SQS**, enabling real-time transaction processing and asynchronous communication.

- Developed responsive front-end components using React.js, JavaScript, HTML5, and CSS3, supporting intuitive enterprise user interfaces and full-stack application workflows.
- Contributed to backlog refinement, participated in code reviews, and collaborated with team members to deliver full-stack features in Agile sprints.
- Implemented CI/CD pipelines using Jenkins, Git, and Maven, enabling automated build, test, and deployment processes across backend services.
- Containerized applications using Docker and supported cloud-based deployment practices for scalable release management.
- Collaborated with product, QA, DevOps, frontend, and business teams to deliver end-to-end enterprise solutions and meet sprint goals.
- Applied AI-assisted development tools to accelerate code generation, documentation, troubleshooting, and developer productivity across engineering workflows.
- Applied **secure coding practices**, encryption standards, and compliance controls to protect sensitive payment and cardholder data.
- Troubleshot and resolved production issues, participated in on-call rotations (Sev2 incidents), and built Python-based operational scripts to support modeled changes, automation, and faster issue remediation.
- Conducted root cause analysis for production and performance issues and implemented durable defect fixes across backend services.
- Designed and implemented **blue-green deployments**, **canary releases**, and **fault-tolerant strategies** to minimize production risks.
- Ensured high code quality through code reviews, unit testing, integration testing, and automated testing using JUnit, Mockito, Selenium, and Postman.
- Utilized JIRA and Git-based workflows to support sprint tracking, code reviews, and collaborative engineering standards.

Environment: Java, Python, Spring Boot, Spring Framework, Spring Security, Spring Cloud, Apache Spark, REST APIs, GraphQL, Microservices, AWS (Glue, Lambda, Step Functions, SNS, SQS, S3, DynamoDB, IAM), Gradle, Maven, ANT, pache Kafka, OAuth 2.0, MySQL, PostgreSQL, MongoDB, Docker, Jenkins CI/CD, Git, Splunk, New Relic, Hibernate, JPA, Spring Batch.

Client: AXA Insurance, Bangalore, India. **Jul 2019 – Aug 2021**
Role: Java Back-End Developer

Responsibilities:
- Collaborated with architecture, risk, and compliance teams to ensure alignment with PCI standards and financial regulatory requirements, enhancing audit readiness and governance compliance for enterprise applications.
- Designed and developed secure, scalable microservices architecture using Java, Spring Boot, Spring Security, and cloud deployment patterns for enterprise applications.
- Built event-driven backend systems using Apache Kafka for high-volume transaction processing and asynchronous communication across distributed cloud services.
- Refactored legacy monolithic applications into domain-driven microservices to improve scalability, maintainability, and cloud migration readiness.
- Implemented CI/CD pipelines using Jenkins, Maven, and release automation tooling, enabling automated builds, testing, and deployments, reducing pre-release defects by 45% and supporting Agile delivery.
- Developed and consumed RESTful APIs using JAX-RS (Jersey) and migrated legacy SOAP (JAX-WS, Apache CXF) services to REST-based services using JSON/XML.
- Worked with relational and NoSQL data platforms, including Oracle- and MongoDB-aligned data patterns, to support secure, scalable enterprise applications.
- Built dynamic, responsive web applications using React.js, JavaScript, HTML5, CSS3, and Bootstrap, improving customer onboarding, usability, and front-end responsiveness.
- Implemented ORM and persistence layers using JPA and iBatis, optimizing database interactions across distributed systems and service-oriented applications.
- Automated operational workflows using Java, Python, and UiPath, reducing manual effort by 10+ hours/week and improving operational efficiency.

- Developed internal dashboards and support utilities using JavaScript, REST APIs, and Python scripting, integrating monitoring and observability tooling for real-time system health, telemetry, and performance visibility.
- Built and managed containerized applications using Docker, improving deployment consistency, portability, and scalable cloud delivery.
- Integrated automated testing frameworks using JUnit, Mockito, and Selenium WebDriver for unit, functional, integration, and UI testing across platforms.
- Managed build automation and dependencies using **Maven**, ensuring consistent and reliable deployments.
- Enforced secure coding practices and implemented role-based authentication and authorization across services in distributed cloud environments.

Environment: Java, Python, Spring Boot, Spring Security, REST APIs, GraphQL, Microservices, Apache Kafka, Cloud-Native Applications, Distributed Systems, JAX-RS, React.js, JPA, Oracle, MongoDB, Jenkins, Docker, Maven, Selenium, Mockito, Cucumber.

Client: Maisa Solutions Pt. Ltd., Hyderabad, India. **Sep 2018 – Jul 2019**
Role: Software Engineer

Responsibilities:

- Developed and executed Spring JDBC batch jobs to process and retrieve customer transactions from Oracle, DB2, Teradata, and MySQL databases, ensuring high-performance data handling and accuracy.
- Implemented the persistence layer using **Hibernate**, leveraging **Annotations**, **HQL**, **Criteria API**, and **Named Queries** to build scalable and maintainable data access solutions.
- Configured and utilized **Hibernate interceptors** to enhance performance, logging, and auditing capabilities within the application.
- Built and managed application components using the **Spring Framework**, ensuring modular architecture and seamless integration across layers.
- Applied **Spring AOP (Aspect-Oriented Programming)** for centralized exception handling, logging, and cross-cutting concerns.
- Leveraged **Spring JDBC Template** to simplify database interactions and improve data persistence efficiency.
- Designed and developed microservices architecture using Spring Boot, enabling scalable and loosely coupled services for enterprise applications.
- Developed and consumed RESTful APIs to facilitate seamless communication between microservices and external systems.
- Integrated Apache Kafka for asynchronous, event-driven communication and real-time data streaming across services.
- Implemented client-side validations using **JavaScript** and facilitated server-side communication using **JSON** for data exchange.
- Built robust web-based enterprise applications using Hibernate 3.0, Spring Framework, and JavaScript-based UI integration, ensuring scalability, maintainability, and smooth front-end/back-end interaction.
- Applied **Inversion of Control (IoC)** and **Dependency Injection (DI)** principles using the **Service Locator design pattern** to improve code modularity and testability.
- Managed asynchronous messaging and system integration using Java Message Service (JMS) for reliable communication between distributed components and backend services.
- Collaborated with stakeholders, QA, and business teams to gather and document requirements for accounting and claims modules, aligning solutions with business and technical needs.

Environment: Java, Spring Boot, Spring Framework, Spring MVC, Hibernate, REST APIs, Microservices, Apache Kafka, JMS, Oracle, DB2, Teradata, MySQL, Selenium, Cucumber.

Education:

- **Master of Science** in **Information Technology** from **University of Cincinnati, USA.**
- **Bachelor of Technology** in **Electronics communication Engineering** from **Vardhaman College of Engineering**, **India**.
